Evolutionary Analysis of the Iterated Symmetric Centipede Game

with All-$M/2$, All-1, and Grim Strategies

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1 Introduction

This report presents the numerical results of an evolutionary game-theoretic analysis of the **Iterated Symmetric Centipede (ISC)** game, based on the work of Lazaridis and Kehagias [1]. The analysis studies the co-evolutionary dynamics of three strategies — All-$M/2$, All-1, and Grim — under both continuous Replicator Dynamics and discrete Markov Chain Dynamics.

The key modification with respect to the original paper is the replacement of the **All-**M strategy with **All-**$M/2$, which cooperates at the midpoint level $\rho_{M/2}$ rather than the maximum ρ_M. This single change alters the payoff matrix, the set of absorbing states, and the equilibrium structure in important ways that are documented throughout this report.

The MATLAB implementation delivers:

- functions `xRepDyn`, `PhasePlot`, `PStateTransitionGraph`, `sMarkDyn`;
- scripts `xRep001` and `xMark001` that reproduce Figures 2–5 of [1] and provide extended parameter sensitivity analysis.

2 The ISC Game Model

2.1 The One-Shot Symmetric Centipede (OSC)

The OSC is a two-player extensive-form game with M stages. At each stage i, the active player may *terminate* (take action ρ_i, ending the game) or *pass* the move to the opponent. The first player to terminate receives a share $p \in (1/2, 1]$ of the joint payoff, while the passive player receives the complementary share $1 - p$.

The resulting payoff matrix A (to the row player) is:

$$A(i,j) = \begin{cases} i & \text{if } i = j \\ (2i+1)\,p & \text{if } i < j \\ (2j+1)\,(1-p) & \text{if } i > j \end{cases}$$

2.2 The Iterated Symmetric Centipede (ISC)

The ISC consists of T repeated plays of the OSC. In each round the first-moving player is selected uniformly at random, so each player moves first in half the rounds on average.

2.3 Strategies

Three ISC strategies are studied (default parameters: $M = 4$, $k = M/2 = 2$):

Index	Symbol	Name	Description
1	$\sigma_{\text{All-}M/2}$	All-$M/2$	Always play $\rho_{M/2}$: cooperates at the mid-level.
2	$\sigma_{\text{All-1}}$	All-1	Always play ρ_1: always defects immediately.
3	σ_G	Grim	Play ρ_M in round 1; if the opponent ever plays ρ_m with $m < M$, switch permanently to ρ_1.

Grim trigger rule. The trigger condition is $m < M$:

- Grim *triggers* against All-$M/2$ (plays $\rho_{M/2}$, and $M/2 < M$).
- Grim *triggers* against All-1 (plays ρ_1, and $1 < M$).
- Grim does *not* trigger against another Grim (plays ρ_M, and $M \not< M$).

Consequently, two Grim players cooperate at the highest level every round, yielding the self-play payoff $C(3,3) = T \cdot M$.

3 The ISC Payoff Matrix C

3.1 Symbolic Form

The entry $C(i,j)$ is the payoff to strategy i when facing strategy j over T rounds of the OSC. The symbolic matrix is:

$$C = \begin{bmatrix} \frac{M}{2}T & 3(1-p)\,T & (M+1)p + (T-1)\,3(1-p) \\ 3p\,T & T & 3p + (T-1) \\ (M+1)(1-p) + (T-1)\,3p & 3(1-p) + (T-1) & T\,M \end{bmatrix}$$

3.2 Entry Derivation

Table 1: Derivation of each $C(i,j)$ entry.

Cell	Matchup	Round 1	Trigger?	Rounds $2 \ldots T$	Formula
$C(1,1)$	All-$M/2$ vs All-$M/2$	$A(k,k)$	No	$A(k,k)$	$T \cdot (M/2)$
$C(1,2)$	All-$M/2$ vs All-1	$A(k,1)$	No	$A(k,1)$	$3(1-p)\,T$
$C(1,3)$	All-$M/2$ vs Grim	$A(k,M)$	Yes	$A(k,1)$	$(M+1)p + (T-1) \cdot 3(1-p)$
$C(2,1)$	All-1 vs All-$M/2$	$A(1,k)$	No	$A(1,k)$	$3p\,T$
$C(2,2)$	All-1 vs All-1	$A(1,1)$	No	$A(1,1)$	T
$C(2,3)$	All-1 vs Grim	$A(1,k)$	Yes	$A(1,1)$	$3p + (T-1)$
$C(3,1)$	Grim vs All-$M/2$	$A(M,k)$	Yes	$A(1,k)$	$(M+1)(1-p) + (T-1) \cdot 3p$
$C(3,2)$	Grim vs All-1	$A(M,1)$	Yes	$A(1,1)$	$3(1-p) + (T-1)$
$C(3,3)$	Grim vs Grim	$A(M,M)$	No	$A(M,M)$	$T \cdot M$

3.3 Numerical Values ($M = 4$, $T = 10$)

Table 2: Numerical payoff matrix C for $p = 3/4$, $M = 4$, $T = 10$.

	vs All-$M/2$	vs All-1	vs Grim
All-$M/2$	20	7.5	10.5
All-1	22.5	10	11.25
Grim	21.5	9.75	40

Table 3: Numerical payoff matrix C for $p = 3/5$, $M = 4$, $T = 10$.

	vs All-$M/2$	vs All-1	vs Grim
All-$M/2$	20	12	13.8
All-1	18	10	10.8
Grim	18.2	10.2	40

Why Grim dominates. The diagonal self-play payoffs are: All-$M/2$: $T \cdot (M/2) = 20$; All-1: $T = 10$; Grim: $T \cdot M = 40$. Grim earns twice the payoff of All-$M/2$ in a homogeneous population because it cooperates at ρ_M (without triggering against other Grims).

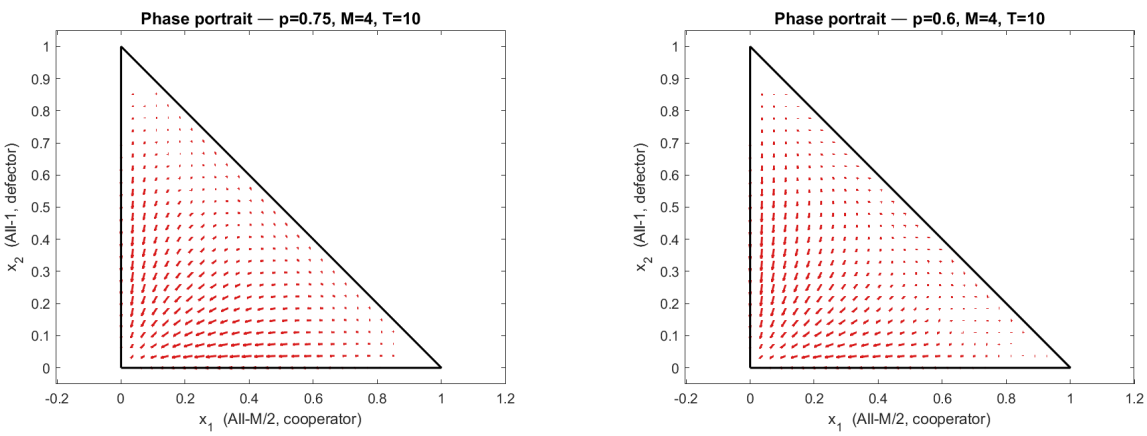
4 Replicator Dynamics

4.1 Model

Let $x = (x_1, x_2, x_3)$ denote the population frequencies of All-$M/2$, All-1, and Grim respectively, with $x_1 + x_2 + x_3 = 1$ and $x_i \geq 0$. The replicator equation is:

$$\dot{x}_m = x_m \left[(Cx)_m - x^\top C x \right], \quad m = 1, 2, 3.$$

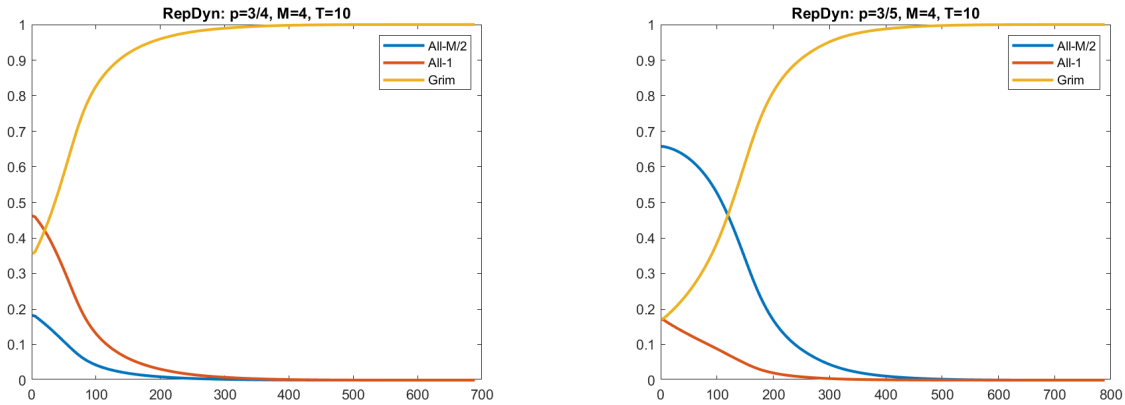
4.2 Phase Portraits (Figures 2–3 of [1])



(a) $p = 3/4$ (b) $p = 3/5$

Figure 1: Phase portraits of the replicator dynamics ($M = 4$, $T = 10$). Axes: x_1 = All-$M/2$ frequency, x_2 = All-1 frequency. The third coordinate $x_3 = 1 - x_1 - x_2$ is implicit.

4.3 Time Series



(a) $p = 3/4$ (b) $p = 3/5$

Figure 2: Replicator dynamics time series ($M = 4$, $T = 10$, random initial condition).

4.4 Rest Points and Stability

The rest points of the replicator dynamics satisfy $(Cx)_m = x^\top Cx$ for all m with $x_m > 0$. The three pure-strategy vertices are always rest points; additional edge and interior equilibria may exist depending on parameters.

Key analytical results from [1] (adapted to All-$M/2$):

- The set $\{(x_1, 0, 1 - x_1) : x_1 \in [0, 1]\}$ (the All-$M/2$–Grim edge, with no defectors) is always a stable equilibrium set; all trajectories converge to it.
- The pure-defector vertex $(0, 1, 0)$ has Jacobian eigenvalues $[-T,\ -(3p-2),\ -T(3p-2)]$ at All-1:
  - $p > 2/3$ (e.g. $p = 3/4$): all eigenvalues negative $\Rightarrow$ $(0, 1, 0)$ is **asymptotically stable**.
  - $p < 2/3$ (e.g. $p = 3/5$): second and third eigenvalues positive $\Rightarrow$ $(0, 1, 0)$ is **unstable**.

The numerical rest point analysis (computed by `RestPoints.m` for all parameter combinations) is saved in `figures/restpoints.txt`.

4.5 Parameter Sensitivity

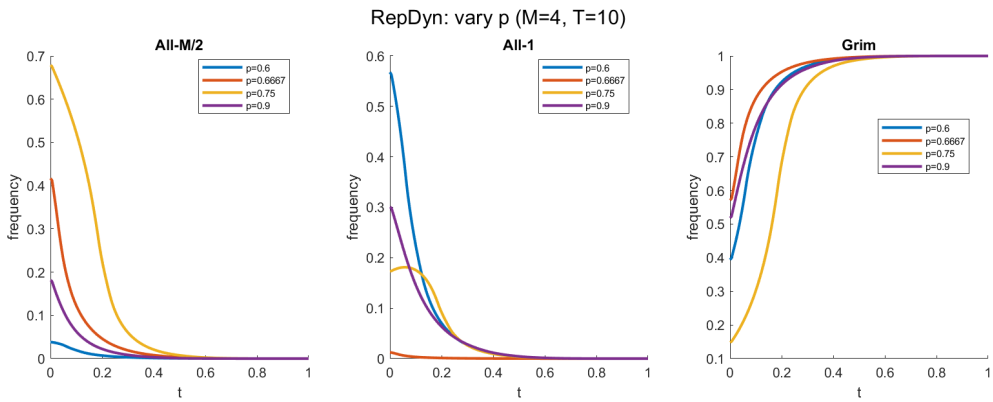


Figure 3: Replicator dynamics: effect of varying $p \in \{3/5, 2/3, 3/4, 9/10\}$ ($M = 4$, $T = 10$). Each subplot shows one strategy's frequency over time.

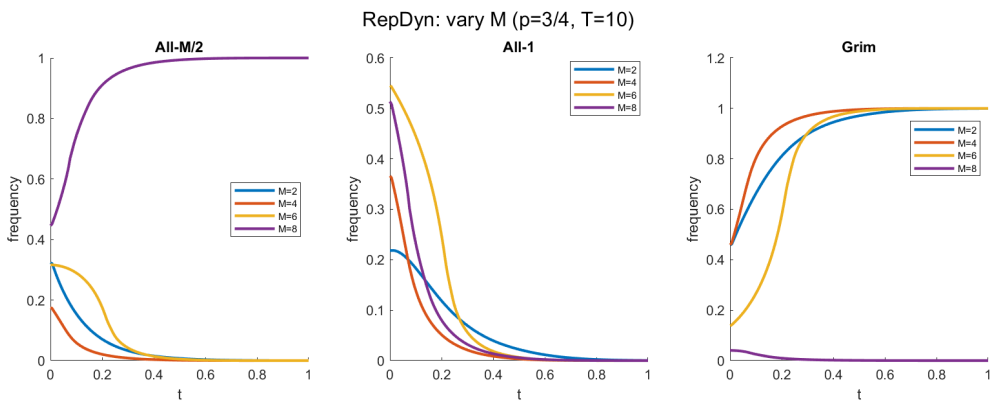


Figure 4: Replicator dynamics: effect of varying $M \in \{2, 4, 6, 8\}$ ($p = 3/4$, $T = 10$).

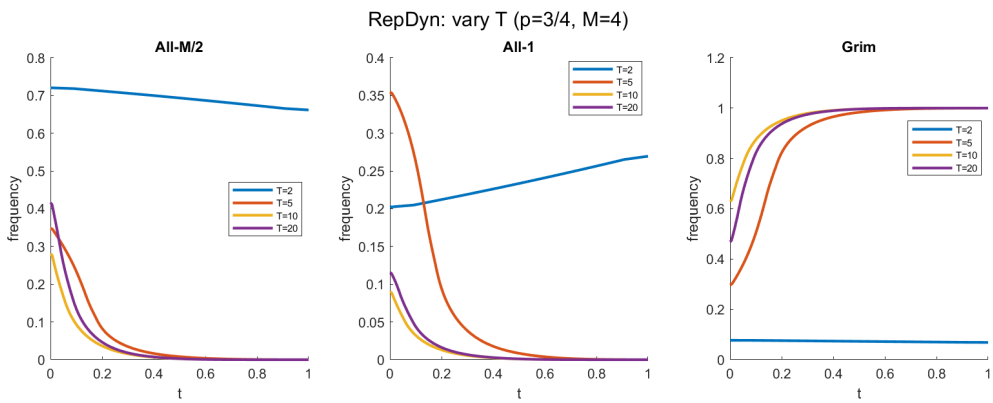


Figure 5: Replicator dynamics: effect of varying $T \in \{2, 5, 10, 20\}$ ($p = 3/4$, $M = 4$).

4.6 Initial Condition Sensitivity

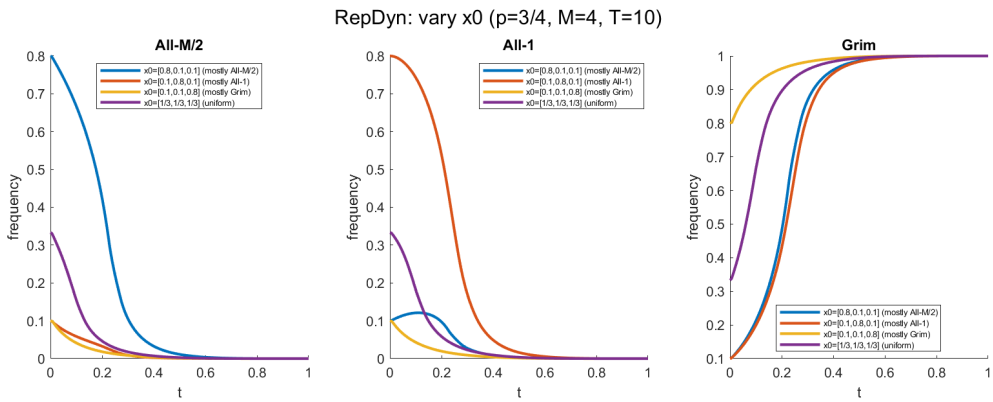


Figure 6: Replicator dynamics: effect of varying initial condition x_0 ($p = 3/4$, $M = 4$, $T = 10$).

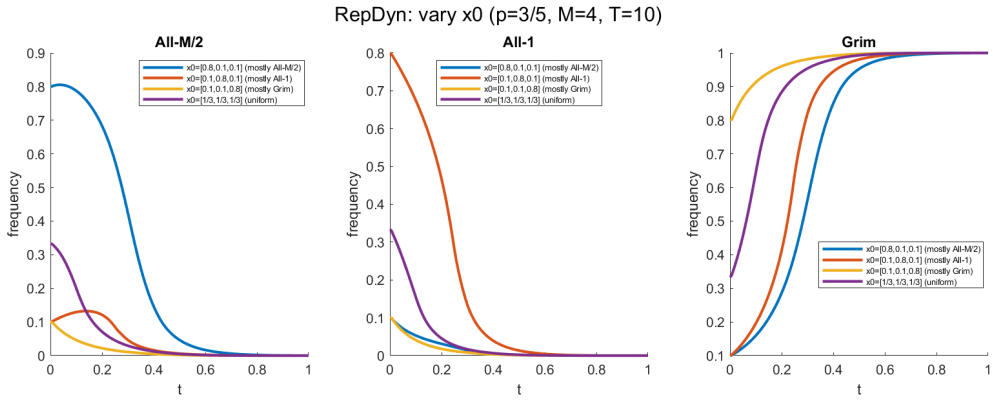


Figure 7: Replicator dynamics: effect of varying initial condition x_0 ($p = 3/5$, $M = 4$, $T = 10$).

5 Markov Chain Dynamics

5.1 Model

The population state is $s = (s_1, s_2, s_3)$ with $s_1 + s_2 + s_3 = N$ and $s_i \geq 0$ integers. There are $(N+1)(N+2)/2$ possible states. The per-player payoff of strategy m in state s is:

$$q_m(s) = (s_m - 1)\, C(m,m) + \sum_{n \neq m} s_n\, C(m,n).$$

Under the **Pairwise Proportional Imitation (PPI)** revision protocol, a randomly selected All-m_1 player switches to All-m_2 with probability:

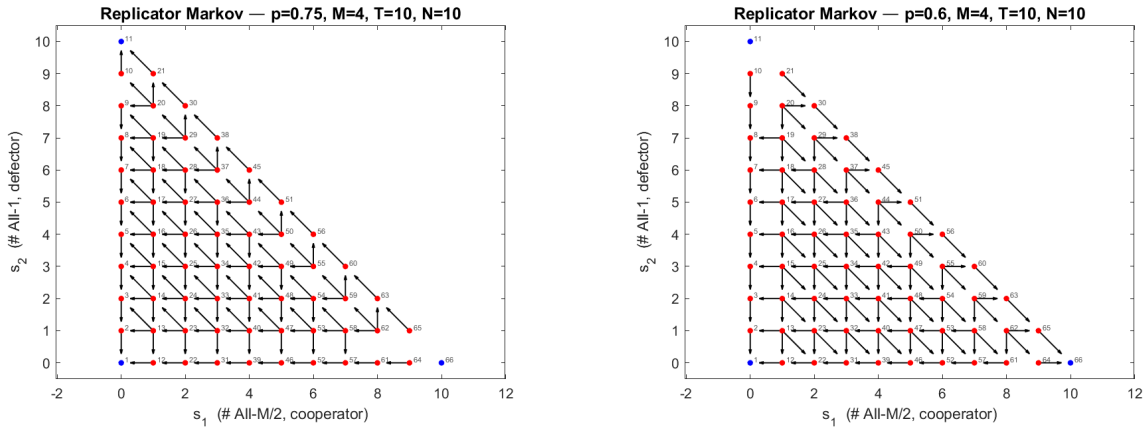
$$\varrho_{m_1 m_2}(s) = \frac{x_{m_2}\,[q_{m_2}(s) - q_{m_1}(s)]_+}{\sum_m x_m\,[q_m(s) - q_{m_1}(s)]_+}.$$

The full transition matrix P is built by `PStateTransitionGraph`.

5.2 Absorbing States

Original paper (All-M): The entire $s_2 = 0$ face is absorbing because All-M and Grim earn the same payoff $T \cdot M$ against each other.
This project (All-$M/2$): $C(1,1) = T \cdot (M/2) = 20 \neq 40 = C(3,3)$, so Grim earns strictly more than All-$M/2$ on the $s_2 = 0$ face. All-$M/2$ players switch to Grim. The **only absorbing states** are the three pure vertices: $(N, 0, 0)$, $(0, N, 0)$, $(0, 0, N)$.

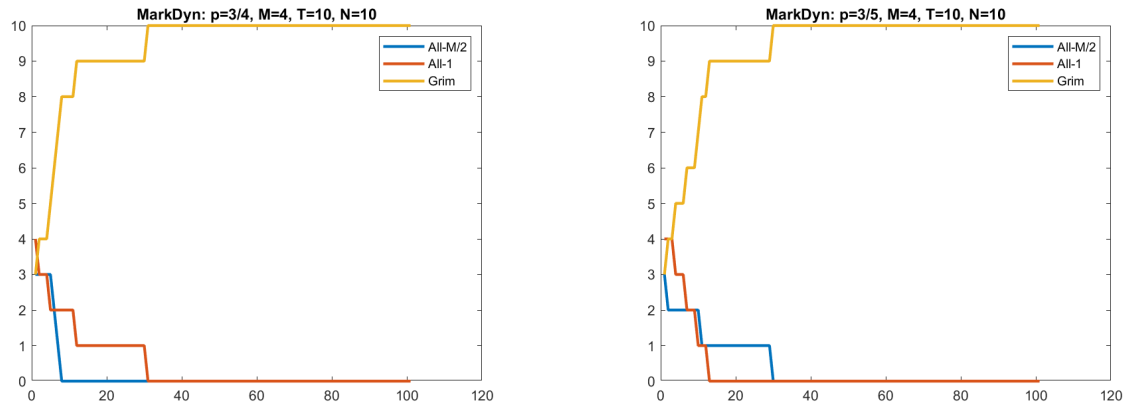
5.3 State Transition Graphs (Figures 4–5 of [1])



(a) $p = 3/4$ (b) $p = 3/5$

Figure 8: Markov state transition graphs ($M = 4$, $T = 10$, $N = 10$). Blue dots: absorbing (pure) states. Red dots: transient states. Arrow thickness indicates transition probability.

5.4 Time Series



(a) $p = 3/4$ (b) $p = 3/5$

Figure 9: Markov dynamics time series ($M = 4$, $T = 10$, $N = 10$, $T_m = 100$, $s_0 = [3, 4, 3]$).

5.5 Parameter Sensitivity

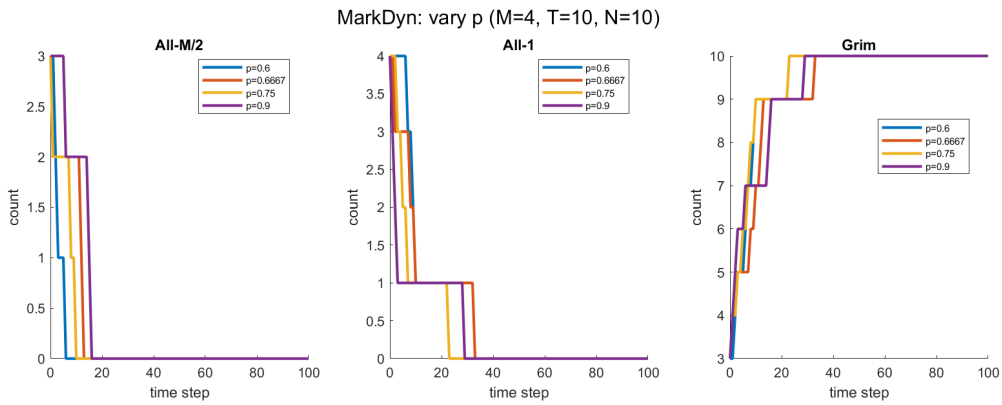


Figure 10: Markov dynamics: effect of varying $p \in \{3/5,\ 2/3,\ 3/4,\ 9/10\}$ ($M = 4$, $T = 10$, $N = 10$).

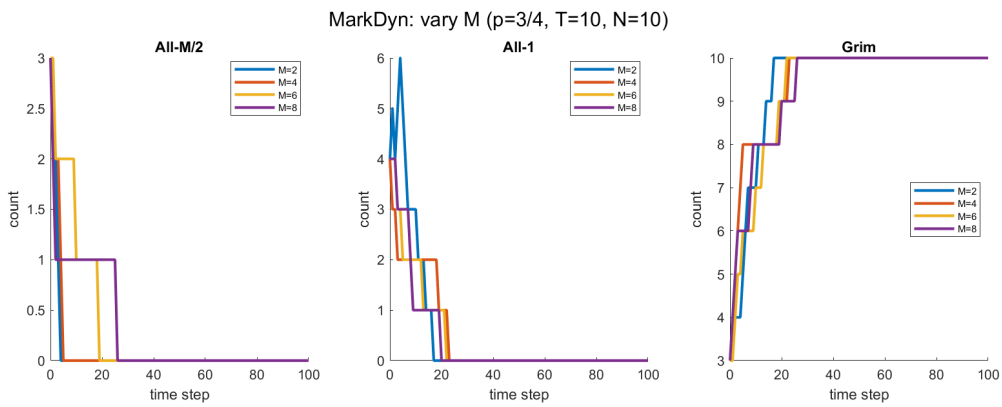


Figure 11: Markov dynamics: effect of varying $M \in \{2, 4, 6, 8\}$ ($p = 3/4$, $T = 10$, $N = 10$).

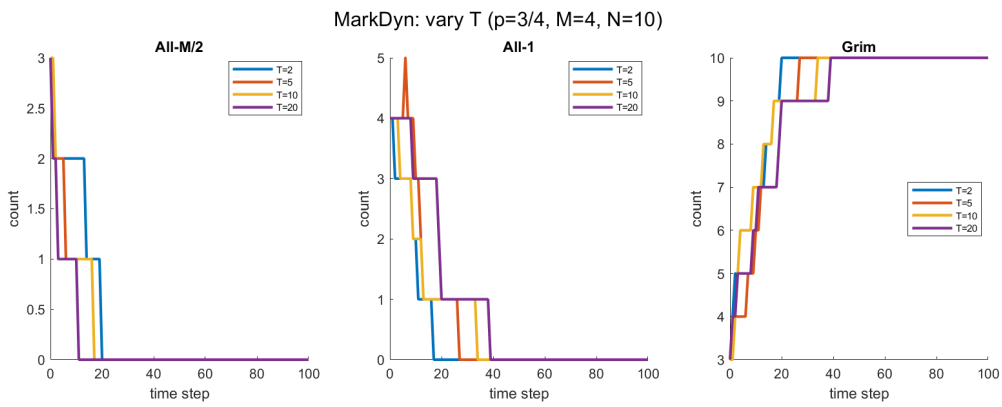


Figure 12: Markov dynamics: effect of varying $T \in \{2, 5, 10, 20\}$ ($p = 3/4$, $M = 4$, $N = 10$).

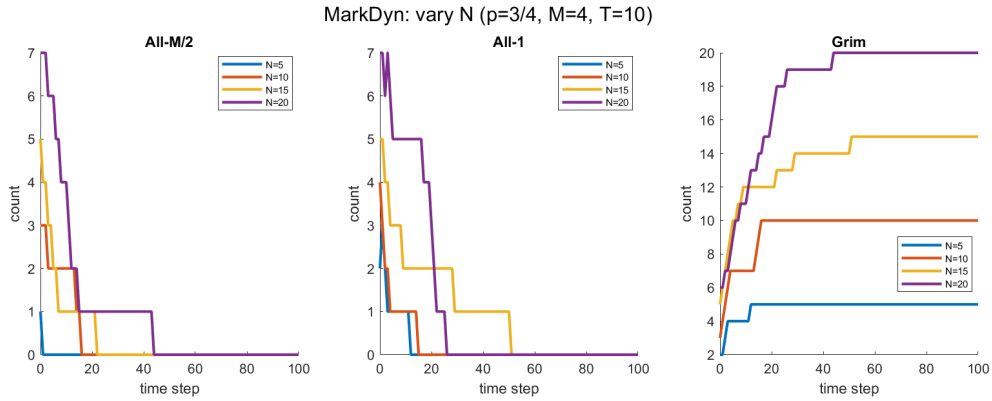


Figure 13: Markov dynamics: effect of varying $N \in \{5, 10, 15, 20\}$ ($p = 3/4$, $M = 4$, $T = 10$).

5.6 Initial State Sensitivity

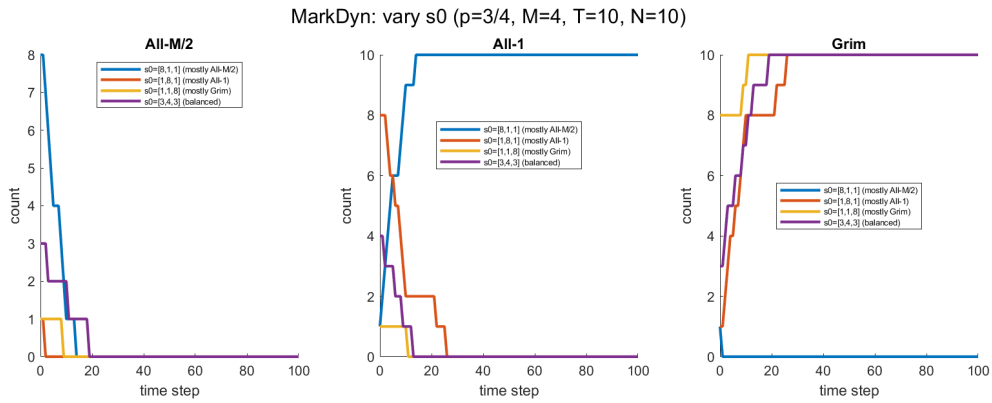


Figure 14: Markov dynamics: effect of varying initial state s_0 ($p = 3/4$, $M = 4$, $T = 10$, $N = 10$).

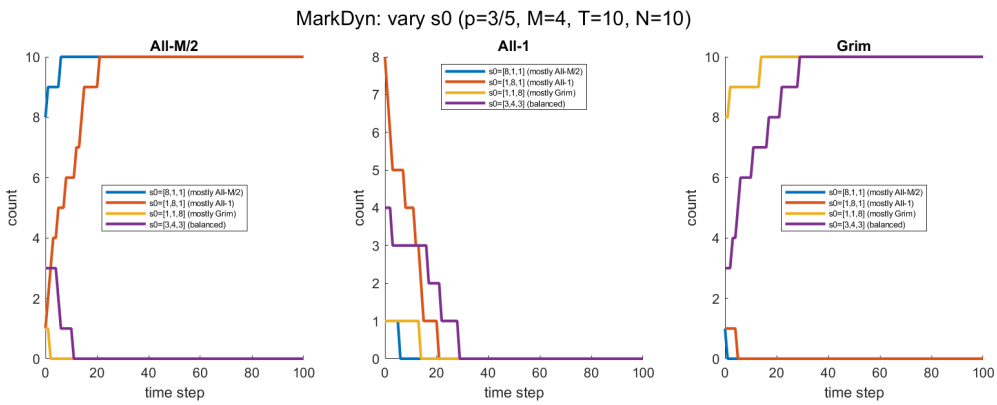


Figure 15: Markov dynamics: effect of varying initial state s_0 ($p = 3/5$, $M = 4$, $T = 10$, $N = 10$).

6 Discussion

6.1 Effect of the All-$M/2$ Modification

Table 4 summarises the key differences between the original paper (which uses All-M) and this project (which uses All-$M/2$).

Table 4: Comparison: original paper (All-M) vs. this project (All-$M/2$).

Aspect	Original (All-M)	This project (All-$M/2$)
Cooperator strategy	Always play ρ_M	Always play $\rho_{M/2}$
$C(1,1)$	$T \cdot M = 40$	$T \cdot (M/2) = 20$
$C(3,3)$	$T \cdot M = 40$	$T \cdot M = 40$ (unchanged)
Grim trigger vs cooperator	Never ($M \not< M$)	Always ($M/2 < M$)
s_2=0 face absorbing?	Yes (entire face)	No (only pure vertices)
Absorbing states	$(0, N, 0)$ + s_2=0 face	$(N, 0, 0)$, $(0, N, 0)$, $(0, 0, N)$

6.2 The $p = 2/3$ Threshold

A critical threshold emerges at $p = 2/3$:

- $p > 2/3$ (e.g. $p = 3/4$): the pure-defector state $(0, 1, 0)$ is asymptotically stable. Both Grim and All-1 can be long-run outcomes.
- $p < 2/3$ (e.g. $p = 3/5$): $(0, 1, 0)$ is unstable. All-1 is driven out; the population converges to the All-$M/2$–Grim edge.

This threshold is visible in the phase portraits (Figure 1) and confirmed by the Markov dynamics (Figure 10).

6.3 Grim Dominance

Grim is the uniquely dominant strategy in self-play ($C(3,3) = 40$, which is twice $C(1,1) = 20$ and four times $C(2,2) = 10$). In the Markov dynamics this is reflected by the strong drift toward the pure-Grim absorbing state $(0, 0, N)$, regardless of the initial condition, especially when $p < 2/3$.

7 Conclusion

This project has reproduced and extended the numerical results of Lazaridis–Kehagias [1] for the ISC game with strategies All-$M/2$, All-1, and Grim. The main findings are:

1. The payoff matrix C is analytically derived and numerically verified for both $p = 3/4$ and $p = 3/5$.
2. Replicator dynamics phase portraits (Figures 2–3 of [1]) confirm the $p = 2/3$ stability threshold for the All-1 equilibrium.
3. Markov state transition graphs (Figures 4–5 of [1]) confirm that, because All-$M/2$ earns less than Grim in self-play, the $s_2 = 0$ face is no longer absorbing: only the three pure states are.

4. Extended parameter sensitivity analysis shows consistent effects: higher p promotes All-1 stability; higher M and T amplify Grim's payoff advantage; initial conditions affect speed of convergence but not the qualitative long-run outcome.

References

[1] I. Lazaridis and A. Kehagias, *Iterated Symmetric Centipede: Learning and Evolutionary Games*, March 2026.